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High Power CW Laser for Co-Packaging

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Abstract: A high-power, Continuous-Wave (CW) laser prototype for Co-Packaged Optics (CPO) applications is demonstrated with more than 10% efficiency. © 2022 The Authors

1. Introduction

The amount of annual global data center traffic has tripled in the last five years, and inside data centers thanks to drivers such as video, AI, and HPC [1]. Power consumption of switches such as 51T and 102T switches might exceed cooling facility limits if traditional techniques with pluggable optics are used. CPO is one of the key initiatives to reduce power generation in datacenter networking systems. CPO-based switch designs utilizing external laser sources, separation of the temperature-sensitive lasers from the switch space limitations on Silicon-Photonic (SiPho) die by eliminating the integrated laser sources, each laser would need to provide approximately 22 to 25dBm of optical power output, it is necessary for the ELS to provide high Power Conversion Efficiency (PCE) to achieve the power reduction sought. Preliminary PCE target of 10% should be achieved.

2. Laser Structure, Device Fabrication and Characterization

The distributed feedback (DFB) laser structure is similar to the one used in [2]. The laser structure in InGaAsP/InP material system features a low QW confinement factor, high saturation power, and large spot size. The far-field divergence angle is small, making fiber coupling possible. A large optical cavity design with a cavity length of 4mm and series resistances and therefore leads to high output power and high PCE. The composition is adjusted to target photoluminescence wavelength that aligns the gain peak at 45°C. The barrier height is chosen to make the lasers relatively temperature insensitive. Based on the coupled-wave method, is used to simulate the single-mode selectivity and the electric field distribution within the cavity for various grating coupling strengths. The optimal grating period is selected based on the simulation results. First-order DFB grating is patterned by high-resolution electron beam lithography.